Module 4: Integration

Ph.D. Math Camp

Antiderivatives

Antiderivatives

- In many problems, the derivative of a function is known, and the goal is to find the function f itself. For example, a sociologist who knows the rate at which the population is growing may wish to use this information to predict future population levels.
- The process of obtaining a function f from its derivative f' is called **antidifferentiation** or **integration**.
- Consider a function f and suppose it is the derivative of another function, say F. Then, $F'(x) = f(x)$ for every x in the domain of f.
- $F(x)$ is said to be an antiderivative of $f(x)$.

Antiderivatives

- A function has more than one antiderivative. For example, one antiderivative of the function $f(x) = 3x^2$ is $F(x) = x^3$ since $F'(x) = 3x^2 = f(x)$.
- But $x^3 + 12$ and $x^3 + e$ are also antiderivatives of $f(x) = 3x^2$.
- If F is an antiderivative of the continuous function f, then any other antiderivative of f differs from F by a constant.
- This is because any two functions with the same derivative (slope) in a domain must have parallel graphs (i.e., differ by a constant).

Fundamental Property of Antiderivatives

- We will represent the family of all antiderivatives of $f(x)$ by using the symbolism

$$\int f(x)dx = F(x) + C$$

  which we call the **indefinite integral** of f. Thus, $F(x) + C$ has derivative $f(x)$. dx, as we saw before, indicates the antidifferentiation process with respect to the variable x.
- Differentiate your answer $F(x) + C$ to check the antidifferentiation calculation. If the derivative equals $f(x)$, your calculation is correct.

Rules for Integrating Common Functions

- Constant Rule: $\int k dx = kx + C$
- Power Rule: $\int x^n dx = \frac{1}{n+1} x^{n+1} + C \quad (\text{for } n \neq -1)$
- Logarithmic Rule: $\int \frac{1}{x} dx = \ln|x| + C$ for all $x \neq 0$
- Exponential Rule: $\int e^{kx} dx = \frac{1}{k} e^{kx} + C$
- Constant multiple Rule: $\int kf(x)dx = k \int f(x)dx$
- Sum Rule: $\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$
- Difference Rule: $\int [f(x) - g(x)]dx = \int f(x)dx - \int g(x)dx$

Integration Examples

1. $\int 4dx$
2. $\int 5x^4 dx$
3. $\int 9e^{-3x} dx$
4. $\int dx$
5. $\int x^{\frac{2}{3}} dx$
6. $\int x^{\frac{-1}{5}} dx$
7. $\int \frac{dx}{x}$
8. $\int \frac{1}{3x} dx$
9. $\int \sqrt{x} dx$
10. $\int \frac{dx}{\sqrt[3]{x}}$
11. $\int (5x^3 + 2x^2 + 3x) dx$
12. $\int (2x^6 - 3x^4) dx$
13. $\int 7xydy$

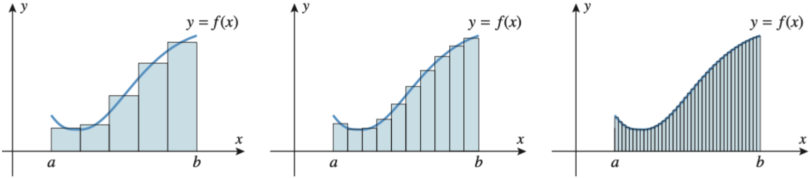
Definite Integration

Definite Integration

- We now revisit the area problem. How do we find the area between the graph of f and the interval $[a, b]$ on the $x-$axis?
- As we did with the tangent line problem, we start with an approximation using what we know.

Definite Integration

- Divide the interval $[a, b]$ into n equal subintervals, and over each subinterval, construct a rectangle that extends from the x-axis to any point on the curve $y = f(x)$ that is above the subinterval.



Definite Integration

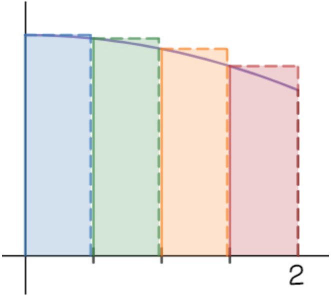
- For each n, the total area of the rectangles can be viewed as an approximation to the exact area under the curve over the interval $[a, b]$.
- Suppose we subdivide $[a, b]$ into four subintervals (rectangles) with the first being $[x_1, x_2]$ and the last being $[x_4, x_5]$
- Since each of these represents a change in x, we may refer to them as $\Delta x_1, \cdots, \Delta x_4$, respectively.
- Each rectangle, at the function, will have a height. For the first rectangle it will be $f(x_1)$ with a width of Δx_1.
- In general, the ith block has a height $f(x_i)$ and a width Δx_i.

Definite Integration

- The total area, A^* of this set of blocks is the sum

$$A^* = \sum_{i=1}^{n} f(x_i)\Delta x_i$$

- This is not the true area but a rough approximation. In the figure below we would be overestimating the area.
- If we try a finer and finer segmentation of the interval $[a, b]$ so that n is increased and Δx_i is shortened indefinitely we will approach the true area A.



Definite Integration

- Carried to the limit, shortening the width of the rectangles yields
$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x_i = \lim_{n\to\infty} A^* = \text{area } A$$
- This is the formal definition for the area under a curve.

Definite Integration

- The definite integral expression summation expression is $\int_a^b f(x)dx$ based on the summation expression $\sum_{i=1}^{n} f(x_i)\,\Delta x_i$
- The replacement of Δx_i by the differential dx is done in the same spirit $\Rightarrow$ express $f(x_i)\Delta x_i$ as $f(x)dx$
- $\int_a^b$ represents the sum from a to b.
- Thus,

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x_i = \text{area} A$$

Definite Integration

- If $f(x)$ is a continuous function and $f(x) \geq 0$ on the interval $a \leq x \leq b$, then the region under the curve $y = f(x)$ above $a \leq x \leq b$ has the area:

  $$A = \int_a^b f(x)dx$$

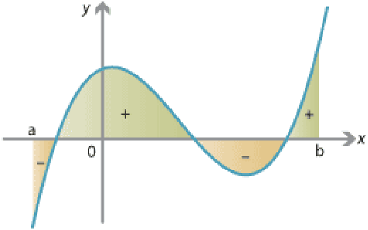
  If $f(x)$ is a continuous function on the interval $a \leq x \leq b$, then

  $$\int_a^b f(x)dx = F(b) - F(a)$$

  where $F(x)$ is any antiderivative of $f(x)$ on $a \leq x \leq b$.
- Thus, to calculate a definite integral, we: 1) figure out an antiderivative F, 2) evaluate at b and a, and then subtract $F(a)$ from $F(b)$.

Definite Integration

- We can calculate the area over or under a curve
- We can have a *negative* area which lies *below* the x-axis and above a given curve.



Properties of Definite Integrals

- The interchange of limits of integration changes the sign of the definite integral

$$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

- We can show this as follows

$$\int_a^b f(x)dx = F(b) - F(a) = -[F(a) - F(b)] = -\int_b^a f(x)dx$$

Properties of Definite Integrals

- A definite integral has a value zero when the two limits of integration are identical

$$\int_a^a f(x)\,dx = 0$$

- This means that the area under any single point is zero. Under our area interpretation, this would mean we have a (one-dimensional) line and not a rectangle.

Properties of Definite Integrals

- A definite integral can be expressed as a sum of a finite number of definite subintegrals as follows:

$$\int_a^d f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx + \int_c^d f(x)dx$$

- Sometimes described as the *additive* property.

Integration

Evaluate the following definite integrals.

1. $\int_1^4 10x\,dx$
2. $\int_1^3 (4x^3 + 6x)dx$
3. $\int_0^5 (2x^3)dx$
4. $\int_1^{64} x^{\frac{-2}{3}}\,dx$
5. $\int_1^2 (x^3 + x + 6)dx$
6. $\int_0^3 4e^{2x}dx$
7. The area of the region under the line $y = 2x + 1$ and above the interval $1 \leq x \leq 3$.

Integration by Parts

Recall the product rule:

$$(f(x)g(x))' = f'(x)g(x) + g'(x)f(x)$$

Integrate both sides

$$\implies f(x)g(x) = \int f'(x)g(x)dx + \int g'(x)f(x)dx$$

Rearrange,

$$\int g'(x)f(x)dx = f(x)g(x) - \int f'(x)g(x)dx$$

Integration by Parts

$$\int g'(x)f(x)dx = f(x)g(x) - \int f'(x)g(x)dx$$

- For example, $\int xe^x \, dx$
- We can let $f(x) = x, g'(x) = e^x, f'(x) = 1, g(x) = e^x$

$$\Rightarrow xe^x - \int 1 \cdot e^x dx = xe^x - e^x + c$$

Integration by Parts

1. $\int \ln x dx$
2. $\int x \ln x dx$
3. $\int x^2 e^{-x} dx$

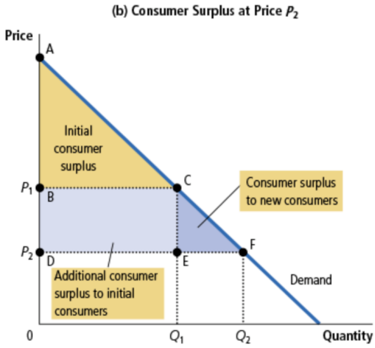
Applications

Consumer Surplus

- CS: A buyer's maximum amount they would be willing to pay for a good is their willingness to pay; a measure of how much that buyer values the good.
  - Suppose you go to buy a car that you have decided, beforehand, that the most you will pay is \$20,000. If the car is more than that you do not buy it.
  - If you find the car at \$15K, then you got a bargain $\rightarrow$ you get a consumer surplus (CS) of 5K. CS is amount a buyer is willing to pay for a good minus the amount buyer actually pays for it.
- For all demand the area below demand and above price measures CS in a market (i.e. the sum of the individual CS of all buyers).

Consumer Surplus

- Consumer surplus is closely related to the demand curve. We can use the market demand curve to get a measure of consumer well-being using CS.



- As $P \downarrow$ consumers are better off. If we lower the price quantity increased and there is additional CS to original consumers (BCED) and to new consumers (CEF). New total CS AFD.

What does CS Measure?

- CS is a pretty good measure of economic well-being because it is the benefits buyers receive from good as the buyers perceive them.
- In some instances, we may choose to ignore CS. For instance, the market for heroin.
- Measuring CS becomes particularly important when considering how different market structures affect consumer well-being.
- Similarly, **Producer Surplus (PS)** is the amount a seller is paid minus the cost of production– benefit to sellers. If the cost is $ 500 (willingness to sell) but the seller does the job for $ 600 the PS is $100.

Consumer Surplus

Consider the following market demand $q_D = (10 - p)^{\frac{1}{2}}$ and market supply expressed as $q_S = \frac{1}{3}p$. Find the equilibrium price and quantity. Find the consumer and producer surplus at the equilibrium price and quantity.

Surplus

- Rewrite q_D, q_S as functions of price and set equal to each other: $p = 10 - q^2$ and $p = 3q$

$$10 - q^2 = 3q \implies q^2 + 3q - 10 = 0 \implies (q+5)(q-2) = 0$$

$$\implies q = 2, q = -5$$

- Since prices and quantities cannot be negative, we will stick with $q = 2 \implies p = 6$.
- Note that the demand has a y-intercept at $p = 10$. To find the consumer surplus, we will need to find the area under the demand but above the equilibrium price.

Surplus

- For CS, find area under the demand up to $q = 2$.

$$\int_0^2 (10 - q^2)\, dq = 10q\Big|_{q=0}^{2} - \frac{q^3}{3}\Big|_{q=0}^{2} = 20 - 0 - \frac{8}{3} + 0 = \frac{52}{3}$$

- Subtract cost to consumer which is area under the price to $q = 2$. That is just 12.
- Thus $CS = \frac{52}{3} - \frac{36}{3} = \frac{16}{3}$
- For PS

$$\int_0^2 6\, dq - \int_0^2 3q\, dq = 6q - \frac{3}{2}q^2\Big|_{q=0}^{2} = 12 - 6 = 6$$

Marginal Function to Total Cost Function

- Given a total cost function, the process of differentiation can yield marginal cost.
- Given that the process of integration is the opposite of differentiation, we can infer the total cost from a given marginal cost.

Marginal Function to Total Cost Function

1. If the MC of a firm is the following $C'(Q) = 2e^{0.2Q}$ and the fixed cost is $FC = 90$, find the total cost function $C(Q)$.

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- Thus, if we set $Q = 0 \implies C(0) = 90$. That is,

$$10e^{0.2(0)} + c = 90 \Rightarrow 10(1) + c = 90 \implies c = 80.$$

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- Thus, the total cost function is

$$C(Q) = 10e^{0.2Q} + 80$$

Given the following marginal-revenue functions and definitize the constant of integration:

1. $R'(Q) = 28Q - e^{0.3Q}$
2. $R'(Q) = Qe^{0.5Q}$